Machine Learning Services on AWS – Introduction

- Presented by : Praveen
- Faculty Advisor : Dr. Victor Govindaswamy

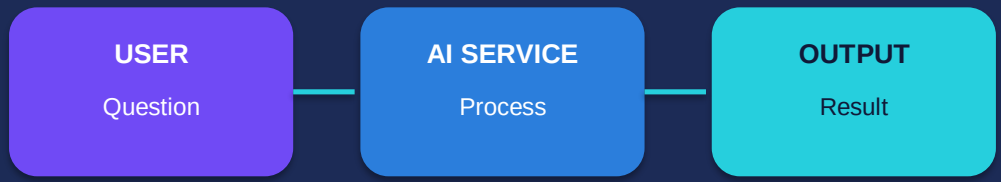
Machine Learning Services on AWS – Introduction

Human review workflows, intelligent search, and the course note about service deprecation

MAIN POINTS

- **AWS machine learning services**
- Amazon Augmented AI (A2I)
- Amazon Kendra
- Amazon Lex
- Amazon Rekognition and Textract

VISUAL CONCEPT | AWS machine learning services overview diagram.



INPUT → PROCESSING → USEFUL RESULT

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

This presentation introduces selected AWS machine learning services from the course section. It focuses on the service purpose, common workflows, and how these tools can support AI-enabled applications.

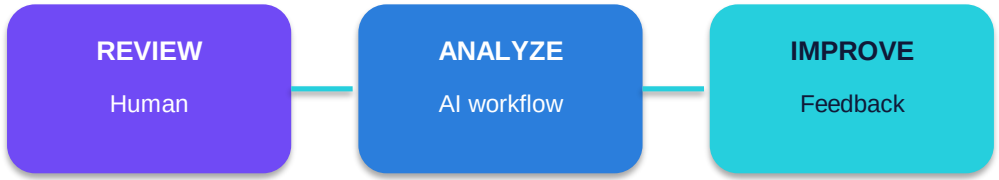
Amazon Augmented AI – Overview

Human review workflows, intelligent search, and the course note about service deprecation

MAIN POINTS

- **Human review workflows**
- Predictions requiring review
- Review tasks
- Confidence thresholds
- Human-in-the-loop AI

VISUAL CONCEPT | Human-in-the-loop AI review workflow.



SERVICE → WORKFLOW → APPLICATION

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

Amazon Augmented AI (A2I) supports human review workflows for eligible machine-learning predictions. Human review can help organizations check selected results and improve confidence in workflows where oversight is required.

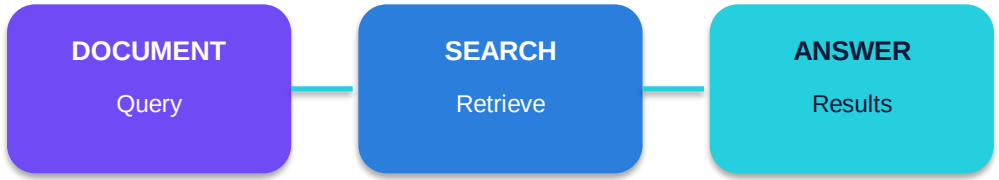
Amazon Augmented AI – Hands-On Workflow

Human review workflows, intelligent search, and the course note about service deprecation

MAIN POINTS

- **Create a review workflow**
- Configure review conditions
- Route selected predictions
- Human reviewer tasks
- Inspect reviewed results

VISUAL CONCEPT | Prediction → review task → human decision workflow.



SERVICE → WORKFLOW → APPLICATION

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

A hands-on workflow helps learners understand how review conditions and human tasks fit into an AI application. Exact steps depend on the AWS service integration and current service availability.

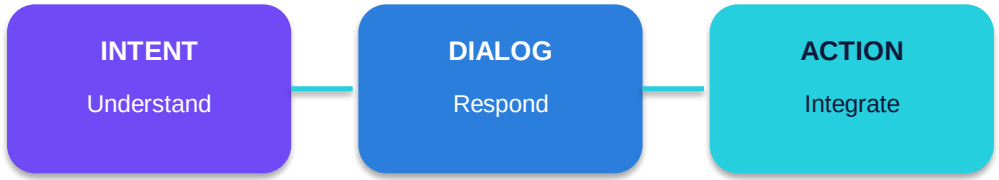
Amazon Kendra – Intelligent Search

Human review workflows, intelligent search, and the course note about service deprecation

MAIN POINTS

- **Enterprise search**
- Natural-language questions
- Relevant content retrieval
- Connectors and data sources
- Search experiences

VISUAL CONCEPT | Enterprise content → search index → relevant results.



SERVICE → WORKFLOW → APPLICATION

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

Amazon Kendra is an intelligent search service designed to help users find relevant information across connected content sources. Learners should review current service documentation and availability before planning new implementations.

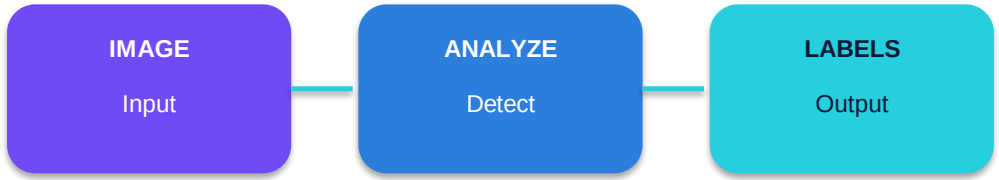
Amazon Kendra – Search Workflow and Use Cases

Human review workflows, intelligent search, and the course note about service deprecation

MAIN POINTS

- **Index content**
- Connect data sources
- Ask questions
- Retrieve relevant passages
- Evaluate search quality

VISUAL CONCEPT | Data sources → index → query → results.



INPUT → SERVICE → OUTPUT

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

A search workflow connects supported content sources to an index, processes user queries, and returns relevant results. Search quality depends on content, permissions, configuration, and evaluation.

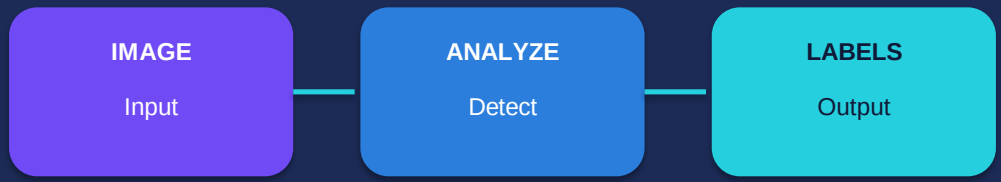
Amazon Lex – Conversational AI

Human review workflows, intelligent search, and the course note about service deprecation

MAIN POINTS

- **Chatbots and voice bots**
- Intent recognition
- Slots and prompts
- Dialog management
- Application integration

VISUAL CONCEPT | Conversational bot dialog flow.



INPUT → SERVICE → OUTPUT

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

Amazon Lex helps developers build conversational interfaces using speech and text. A bot commonly interprets an intent, collects required information, manages the dialog, and invokes application logic when configured.

Amazon Lex – Hands-On Bot Design

Human review workflows, intelligent search, and the course note about service deprecation

MAIN POINTS

- **Define intents**
- Add sample utterances
- Configure slots
- Test dialog paths
- Connect fulfillment logic

VISUAL CONCEPT | Intent → slots → fulfillment → response.



INPUT → SERVICE → OUTPUT

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

A practical Lex exercise demonstrates how to configure a bot and test its conversational behavior. Learners should check the bot's response for expected, missing, and ambiguous inputs.

Amazon Rekognition – Image and Video Analysis

Human review workflows, intelligent search, and the course note about service deprecation

MAIN POINTS

- **Image analysis**
- Object and scene labels
- Face-related analysis capabilities
- Video analysis options
- Responsible use

VISUAL CONCEPT | Media input → analysis → labels or detections.



INPUT → SERVICE → OUTPUT

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

Amazon Rekognition provides image and video analysis capabilities. The exact features available depend on the API and Region; use these tools responsibly and follow applicable privacy, consent, and organizational policies.

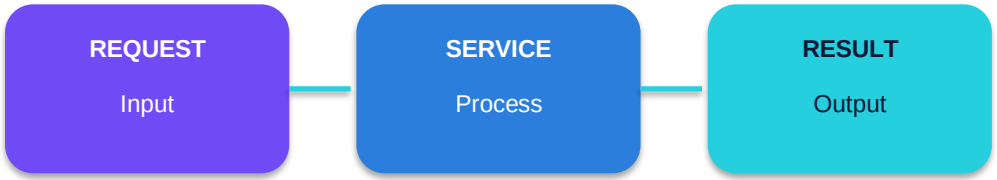
Amazon Rekognition – Hands-On Analysis

Human review workflows, intelligent search, and the course note about service deprecation

MAIN POINTS

- **Submit sample media**
- Run a supported analysis
- Inspect labels or detections
- Review confidence scores
- Validate results

VISUAL CONCEPT | Image analysis request and returned labels.



INPUT → SERVICE → OUTPUT

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

Hands-on practice helps learners see how a supported Rekognition operation processes media and returns structured results. Outputs should be checked against the task and should not be treated as infallible.

Comparing AWS AI Services

Human review workflows, intelligent search, and the course note about service deprecation

MAIN POINTS

- **Augmented AI: human review**
- Kendra: intelligent search
- Lex: conversational interfaces
- Rekognition: image and video
- Textract: document extraction

VISUAL CONCEPT | Service-to-use-case comparison matrix.



INPUT → SERVICE → OUTPUT

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

These services address different tasks rather than serving as interchangeable products. Select a service based on the input data, expected output, workflow, governance requirements, and current AWS service availability.

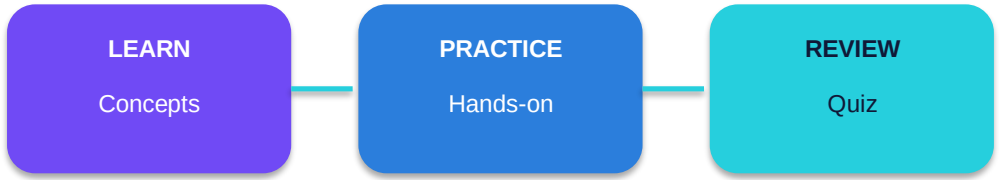
Selecting the Right Service for a Use Case

Human review workflows, intelligent search, and the course note about service deprecation

MAIN POINTS

- **Define the business problem**
- Identify input and output
- Check supported features
- Consider data protection
- Test with representative samples

VISUAL CONCEPT | Use case → service fit → prototype → evaluation.



INPUT → SERVICE → OUTPUT

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

A structured selection process starts with the use case and its constraints. A small evaluation using representative data can help determine whether a service is appropriate before a larger deployment.

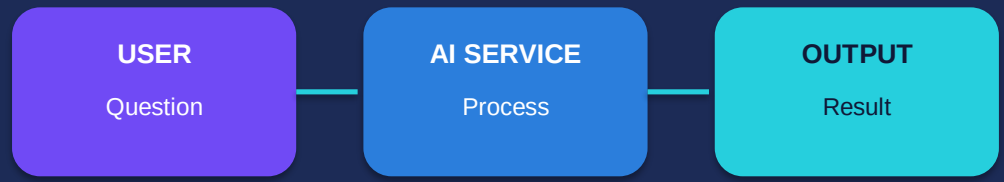
Summary – Human Review, Search and Conversational AI

Human review workflows, intelligent search, and the course note about service deprecation

MAIN POINTS

- **Understand each service purpose**
- Recognize typical workflows
- Practice with hands-on examples
- Validate outputs
- Check current service documentation

VISUAL CONCEPT | AWS AI service learning roadmap.



INPUT → PROCESSING → USEFUL RESULT

Conceptual workflow illustration • not an AWS console screenshot

DESCRIPTION

This presentation introduced Augmented AI, Kendra, Lex, Rekognition, and their typical workflows. The key takeaway is to match each service to a specific problem and validate results before production use.